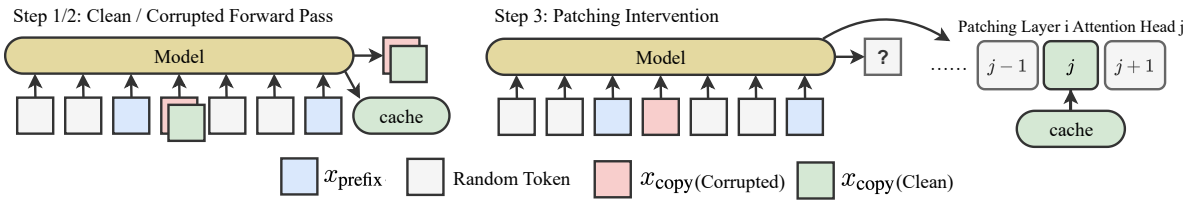


Figure 2: Overall pipeline of the three steps to detect the induction heads in LLMs.

head $(i, j)$ is quantified by the logit recovery:

$$I_{(i,j)} = \frac{\left(\ell^{(i,j)}_{\text{patched}}(x_{\text{copy}}) - \ell_{\text{corrupt}}(x_{\text{copy}})\right)}{\left(\ell_{\text{clean}}(x_{\text{copy}}) - \ell_{\text{corrupt}}(x_{\text{copy}})\right)} \quad (1)$$

where $x_{\text{copy}}$ is the correct continuation token. This double difference formulation controls for baseline noise while measuring the specific contribution of each head.

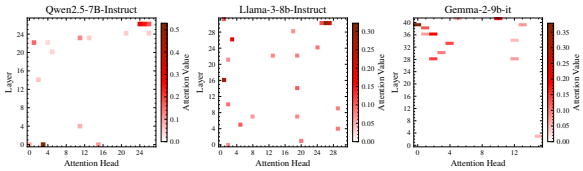


Figure 3: Causal importance $I_{(i,j)}$ of induction head in Qwen2.5-7B-Instruct, Llama-3-8b-Instruct and Gemma-2-9B-it.

**Induction Head Identification** Through systematic activation patching across all model layers, we identify heads with maximal $I_{(i,j)}$ scores. The top $p\%$ of heads by this metric exhibit characteristic induction behavior: strong prefix matching and copy boosting. These heads form a distributed circuit whose combined activity mediates the model's pattern replication capability. Our experiments (See Figure 3) reveal that induction heads typically concentrate in middle and deep layers, with their output projections aligning to the token embedding space. We select $p = 2$ *i.e.,* top $2\%$ heads as induction heads based on the previous works (Olsson et al., 2022; Crosbie and Shutova, 2024).

### 3.2 The Toxicity Theory of Induction Head

**Formalizing Induction Head Dominance.** Let $\mathcal{H}$ denote the set of all attention heads in a transformer-based LLM. We partition $\mathcal{H}$ into induction heads $\mathcal{H}_{\text{ind}} \subset \mathcal{H}$ and non-induction heads $\mathcal{H}_{\setminus\text{ind}} = \mathcal{H} \setminus \mathcal{H}_{\text{ind}}$. We propose the following definitions:

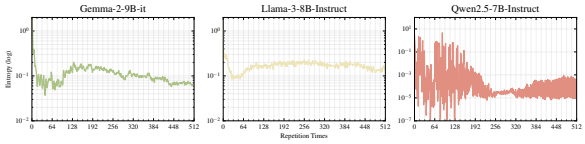


Figure 4: Entropy of outputs when the repetition times increase from 2 to 512.

**Definition 3.2** (Induction Head Toxicity Ratio)**.** For any generation step $t$, define the *induction head toxicity ratio* $\tau_t$ as:

$$\tau_t = \frac{\sum_{(i,j)\in\mathcal{H}_{\text{ind}}} |I^{(t)}_{(i,j)}|}{\sum_{(k,l)\in\mathcal{H}} |I^{(t)}_{(k,l)}|}, \quad (2)$$

where $I^{(t)}_{(i,j)}$ is the causal influence score from Equation 1 for head $(i, j)$ at position $t$.

**Definition 3.3** (Induction Head Toxicity)**.** We define the model experiences *induction head toxicity* at step $t$ if $\tau_t \geq \gamma$, where $\gamma$ is a toxicity threshold [2].

**Lemma 3.4** (Toxicity Propagation)**.** *Once toxicity occurs ($\tau_t \geq \gamma$), subsequent steps satisfy $\mathbb{E}[\tau_{t+1}] \geq \tau_t$ under the model's autoregressive generation process.*

*Proof.* (Informal) Induction heads specialize in pattern completion. When dominant, their output logits $l_t$ bias the next token distribution $p_{t+1}$ toward continuing existing patterns. This creates a self-reinforcing cycle where $p_{t+1}$'s entropy decreases, making induction head strategies increasingly optimal for subsequent predictions. □

**Entropy Reduction Under Toxicity.** The reduction of entropy explains the final repetition of LLM's outputs. Let $H(p_t)$ denote the entropy of the token distribution at step $t$. Define the *toxicity gradient* $\nabla H_t = H(p_t) - H(p_{t+1})$.

[2] We selected a ratio of 0.65 for our experiments.

**Lemma 3.5** (Entropy Collapse)**.** *Under induction head toxicity ($\tau_t \geq \gamma$), the entropy gradient satisfies:*

$$\mathbb{E}[\nabla H_t] \geq k \cdot \tau_t \cdot \log |\mathcal{V}|, \tag{3}$$

*where $k > 0$ is a constant and $\mathcal{V}$ is the vocabulary. This produces exponential entropy decay over successive toxic steps.*

*Proof.* Let the total logits at step $t$ be:

$$\boldsymbol{l}_t = \underbrace{\sum_{(i,j)\in\mathcal{H}_{\text{ind}}} \boldsymbol{l}_{\text{ind}}^{(i,j)}}_{\text{Induction head contributions}} + \underbrace{\sum_{(k,l)\in\mathcal{H}_{\setminus\text{ind}}} \boldsymbol{l}_{\text{other}}^{(k,l)}}_{\text{Non-induction contributions}}.$$

The toxicity ratio $\tau_t$ measures the relative influence of induction heads:

$$\tau_t \propto \frac{\|\boldsymbol{l}_{\text{ind}}\|}{\|\boldsymbol{l}_{\text{ind}}\| + \|\boldsymbol{l}_{\text{other}}\|}, \tag{4}$$

where $\|\cdot\|$ denotes the $\ell_2$-norm. Under toxicity ($\tau_t \geq \gamma$), induction head logits dominate:

$$\|\boldsymbol{l}_{\text{ind}}\| \gg \|\boldsymbol{l}_{\text{other}}\|. \tag{5}$$

The next-token distribution is:

$$p_{t+1}(x) = \text{Softmax}(\boldsymbol{l}_t)_x \propto \exp\left(\boldsymbol{l}_{\text{ind},x} + \boldsymbol{l}_{\text{other},x}\right). \tag{6}$$

Under toxicity, $\boldsymbol{l}_{\text{ind},x}$ dominates for tokens $x$ matching the repetitive pattern. Let $x^*$ be the token maximally boosted by induction heads. For simplicity, assume:

$$p_{t+1}(x^*) \approx 1 - \epsilon, \quad \epsilon \ll 1. \tag{7}$$

The entropy then becomes:

$$H(p_{t+1}) \approx -(1-\epsilon)\log(1-\epsilon) - \sum_{x\neq x^*} \epsilon_x \log \epsilon_x, \tag{8}$$

where $\epsilon_x = p_{t+1}(x)$. For small $\epsilon$, this approximates to:

$$H(p_{t+1}) \approx \epsilon \log |\mathcal{V}|. \tag{9}$$

The entropy gradient is:

$$\begin{aligned}\nabla H_t = H(p_t) - H(p_{t+1}) &\geq \log|\mathcal{V}| - \epsilon \log|\mathcal{V}| \\ &= (1-\epsilon)\log|\mathcal{V}|.\end{aligned} \tag{10}$$

Under toxicity, $\epsilon \leq 1 - \tau_t$ (induction heads suppress non-pattern tokens). Thus:

$$\mathbb{E}[\nabla H_t] \geq \tau_t \cdot \log|\mathcal{V}|. \tag{11}$$

For arbitrary patterns (not just single-token repetitions), the worst-case entropy reduction scales as $k \cdot \tau_t \cdot \log|\mathcal{V}|$, where $k \in (0, 1]$ depends on the pattern length and induction head efficiency. □

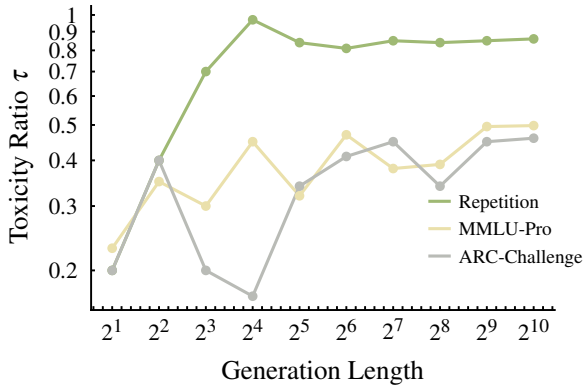


Figure 5: Empirical validation of toxicity propagation dynamics.

### 3.3 Empirical Validation

We validate these theoretical constructs through controlled experiments on Qwen2.5-7B-Instruct. First, we examine the induction head toxicity ratio ($\tau_t$) in two scenarios: a repetition curse setting ( repetition-512) and a general-purpose benchmark ( MMLU-Pro, ARC-Challenge). We track the induction head toxicity ratio over 512 newly generated tokens. The results show that $\tau_t$ increases sharply in repetition-curse scenarios, while it remains relatively low in normal tasks. Additionally, we investigate the entropy collapse phenomenon. Table 1 illustrates the entropy collapse under induced toxicity conditions, and Figure 4 provides a dynamic visualization of this issue.

| Condition | Initial Entropy | Decay Rate |
|---|---|---|
| Normal ($\tau < 0.5$) | 5.2 | 0.3 |
| Toxic ($\tau \geq 0.65$) | 4.8 | 1.7 |

Table 1: Entropy collapse under toxicity conditions (Qwen2.5-7B-Instruct).

## 4 Solution: Induction Head Descaling

Building on our Induction Head Toxicity Theory, we propose a targeted mitigation that dynamically scales down induction head contributions during inference. As illustrated in Figure 6, while induction heads are essential for in-context learning, their increasing influence in repetitive contexts drives toxic behavior. Our position-dependent scaling preserves their normal functionality while reducing their dominance during repetition.

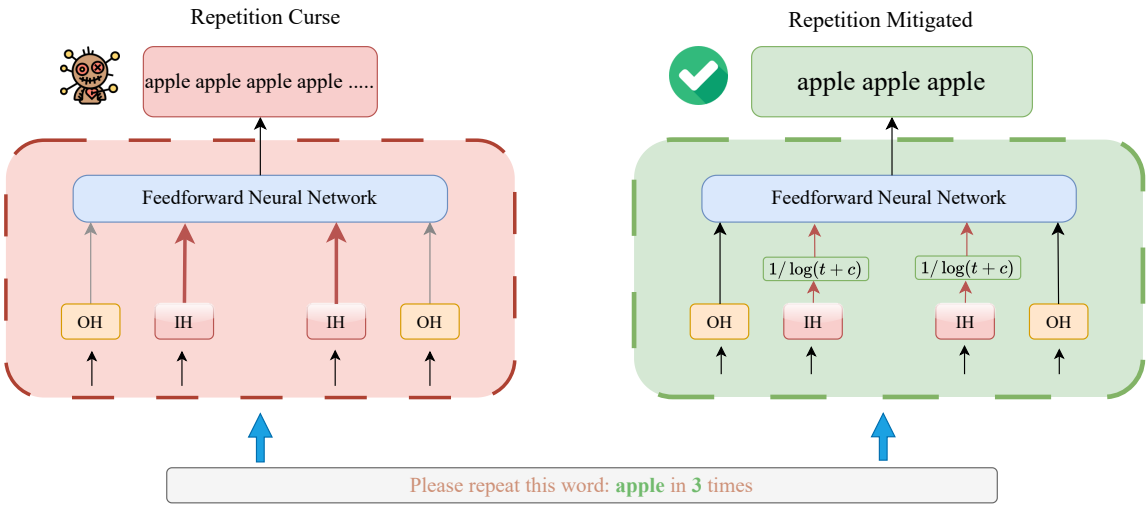


Figure 6: Mechanistic illustration of Induction Head (IH) descaling for mitigating the repetition curse. (Left) Uncontrolled generation where dominant IHs precipitate repetitive output. (Right) Application of a position-dependent scaling factor (e.g., $1/\log(t+c)$) to IH contributions, which attenuates their dominance and restores controlled generation. OH denotes Other Heads.

### 4.1 Logarithmic Output Descaling

Let $\mathcal{H}_{\text{ind}}$ denote the set of identified induction heads. For each head $(i,j) \in \mathcal{H}_{\text{ind}}$ at generation step $t$, we modify its output projection as:

$$\boldsymbol{O}_t^{(i,j)} = \frac{\text{Attn}^{(i,j)}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) \cdot \boldsymbol{W}_V^{(i,j)}}{\log(t+c)}, \quad (12)$$

where $c > 1$ is a stability constant (we use $c = 2$ empirically), and $\log(t+c)$ grows with sequence position. Non-induction heads remain unmodified. This creates a decaying weight profile for induction heads as generation progresses.

**Theorem 4.1** (Toxicity Mitigation)**.** *Let $\tau_t$ be the original toxicity ratio and $\tilde{\tau}_t$ the modified ratio under descaling. Then:*

$$\tilde{\tau}_t \leq \tau_t. \quad (13)$$

*Proof.* The scaled induction head influences become $I_{(i,j)}/\log(t+c)$. This proportional scaling is reasoned because when induction heads are dominant, their outputs $\boldsymbol{O}_t^{(i,j)}$ significantly direct the computation through specific circuits impacting the final logits. Therefore, scaling these already influential outputs proportionally scales their measured importance $I_(i,j)$ due to the linear nature of these circuit pathways. Then:

$$\begin{aligned}\tilde{\tau}_t &= \frac{I_{\text{ind}}/\log(t+c)}{I_{\text{total}} - I_{\text{ind}} + I_{\text{ind}}/\log(t+c)} \\ &= \frac{\tau_t}{(1-\tau_t)\log(t+c) + \tau_t} \leq \tau_t,\end{aligned} \quad (14)$$

where $I_{\text{ind}} = \sum_{(i,j)\in\mathcal{H}_{\text{ind}}} |I_{(i,j)}|$ and $I_{\text{total}} = \sum_{(k,l)\in\mathcal{H}} |I_{(k,l)}|$. □

### 4.2 Mechanistic Interpretation

The logarithmic descaling counteracts two key aspects of induction head toxicity:

**Positional Feedback Loop.** As repetition continues, induction heads normally receive stronger positional cues to reinforce patterns. The $\log(t+c)$ denominator dampens this positive feedback.

**Entropy Preservation.** From Lemma 3.5, entropy collapse scales with $\tau_t$. By bounding $\tilde{\tau}_t \leq \tau_t$, we ensure:

$$\mathbb{E}[\nabla H_t] \leq c \cdot \tau_t \cdot \log|\mathcal{V}|, \quad (15)$$

that slowing entropy decay preserves token diversity over longer sequences.

### 4.3 Implementation Considerations

Our method requires two practical modifications:

1. **Induction Head Identification**: Use the causal patching method from Section 3.1 to detect $\mathcal{H}_{\text{ind}}$ once per model. This can be done with synthetic copy tasks requiring acceptable GPU hour.
2. **Dynamic Scaling During Generation**: Track the generation step $t$ and apply the $\log(t+c)$ scaling factor only to identified induction heads. This adds negligible computational overhead.